UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

3889746187

PHYSICS **5054/04**

Paper 4 Alternative to Practical **October/November 2007**

1 hour

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
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Total	

This document consists of **9** printed pages and **3** blank pages.

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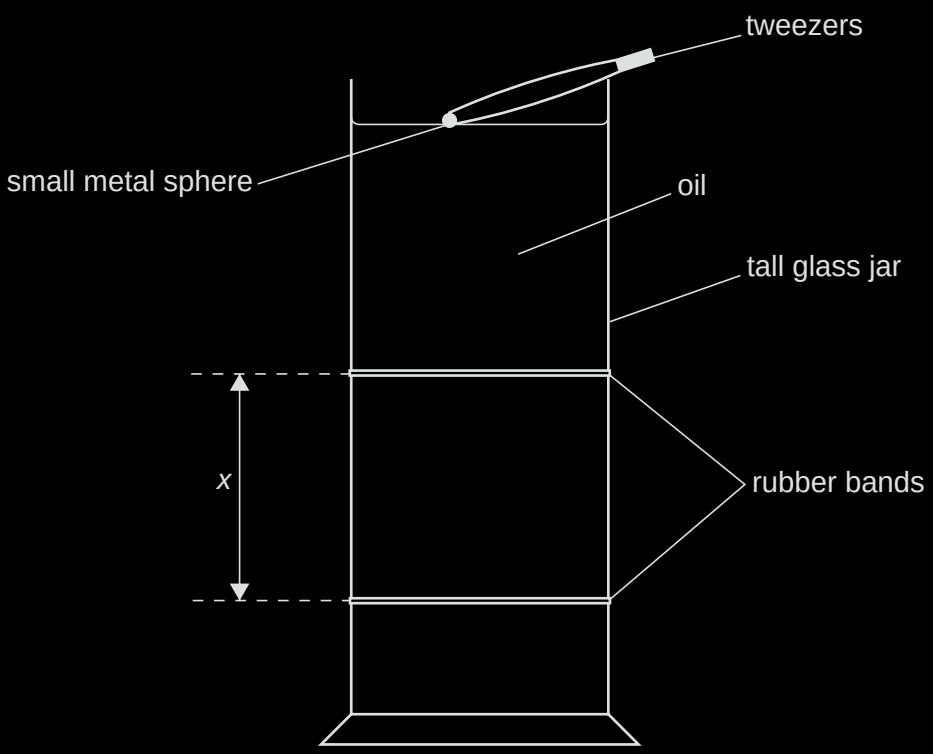


Fig. 1.1

Tweezers hold the small metal sphere below the surface of the liquid before it is released. As the sphere falls, it reaches terminal velocity.

To measure the steady velocity of the sphere, a student measures the time *t* it takes to fall a distance *x*.

(a) Suggest a reason why the student does **not** measure *x* from the surface of the oil.

.....

..... [1]

(b) The student places two rubber bands around the jar as shown in Fig. 1.1.

Explain how the rubber bands make measuring the time more accurate.

.....

..... [1]

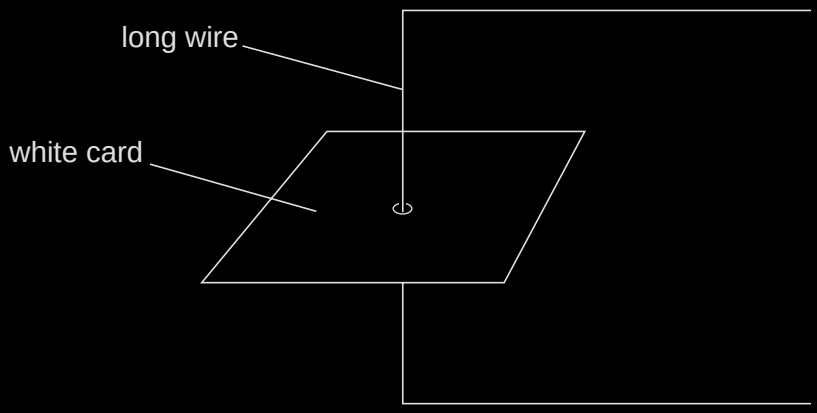


Fig. 2.1

(a) Complete Fig. 2.1 to show a circuit that provides a current in the wire. The circuit should allow the current to be varied and measured. [2]

(b) The student sprinkles iron filings on the card to show the shape of the magnetic field.

Give a reason for each of the following.

(i) The current needs to be large.

.....

..... [1]

(ii) The student must tap the card.

.....

..... [1]

(iii) The student uses small iron filings.

.....

..... [1]

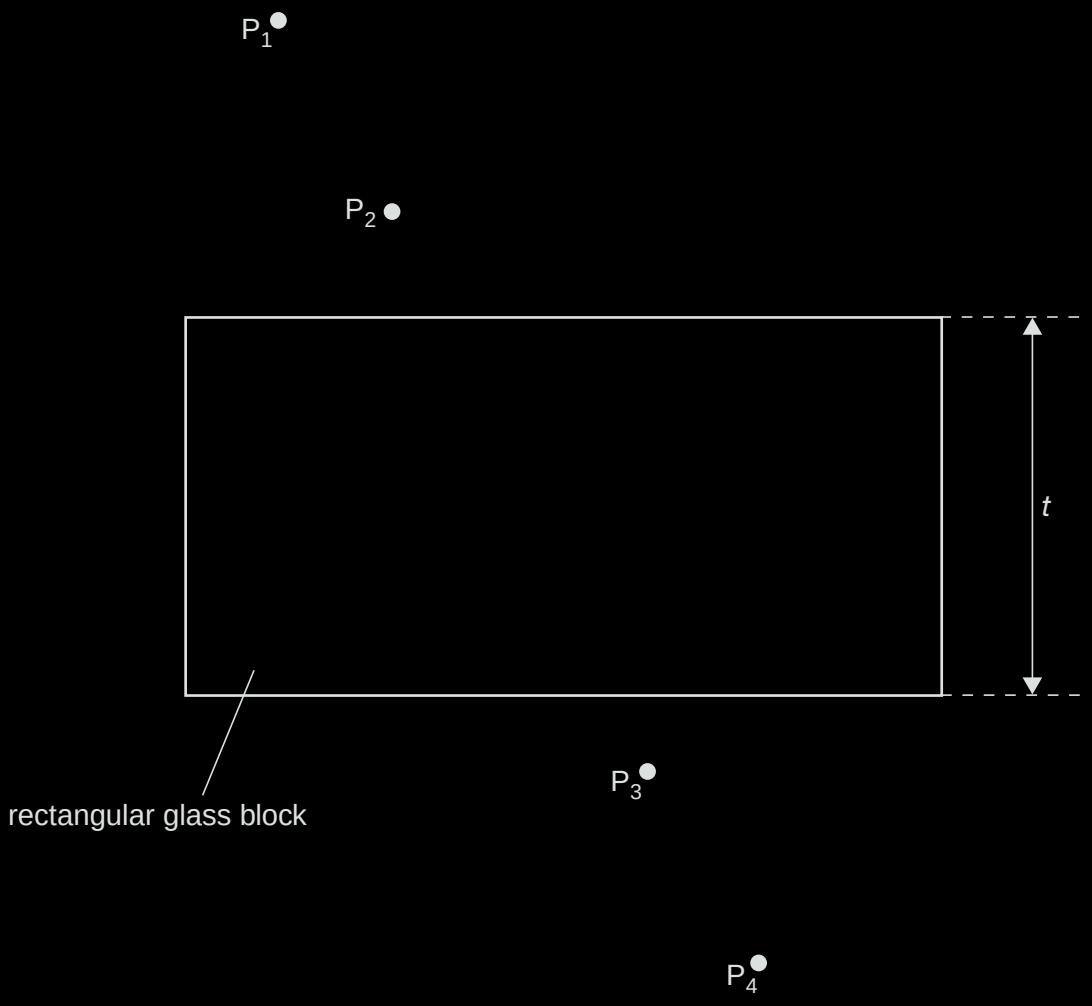


Fig. 3.1

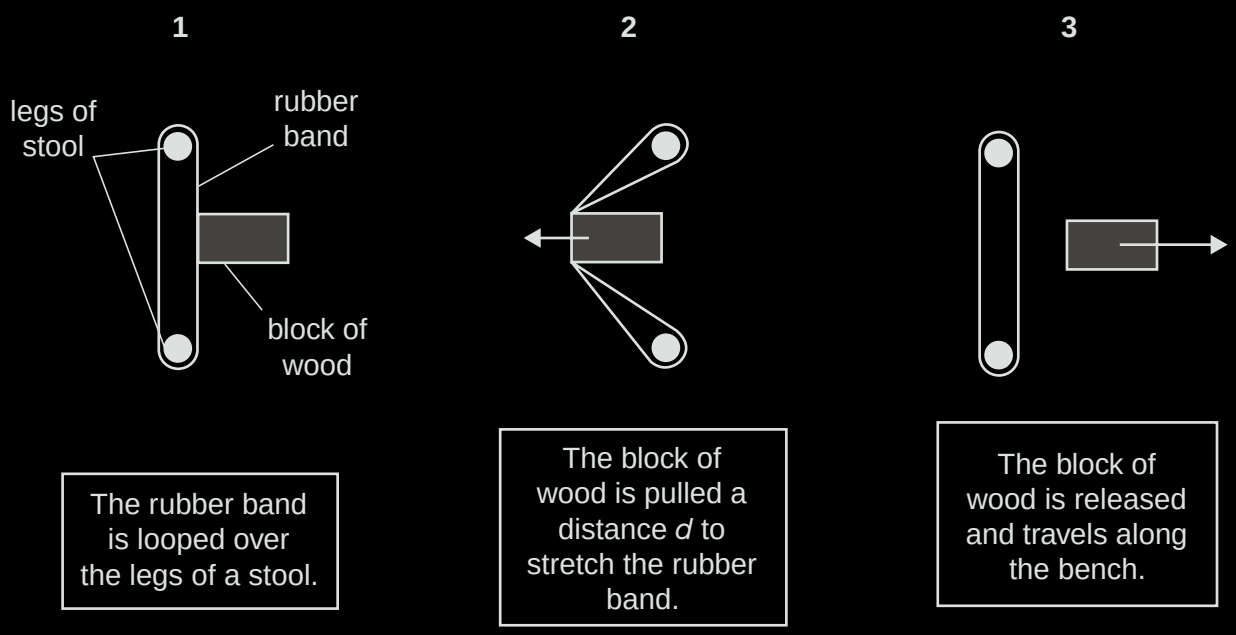


Fig. 4.1

The distance *d* is varied and the distance *D* travelled by the block along the bench is measured for each value of *d*.

The student records the results in a table.

distance *d* / cm	distance *D* / cm
5.0	13.3
6.0	20.9
7.0	28.7
8.0	39.6
9.0	51.8
10.0	62.5

(a) Explain why $D = 0$ when $d = 0$.

.....

..... [1]

